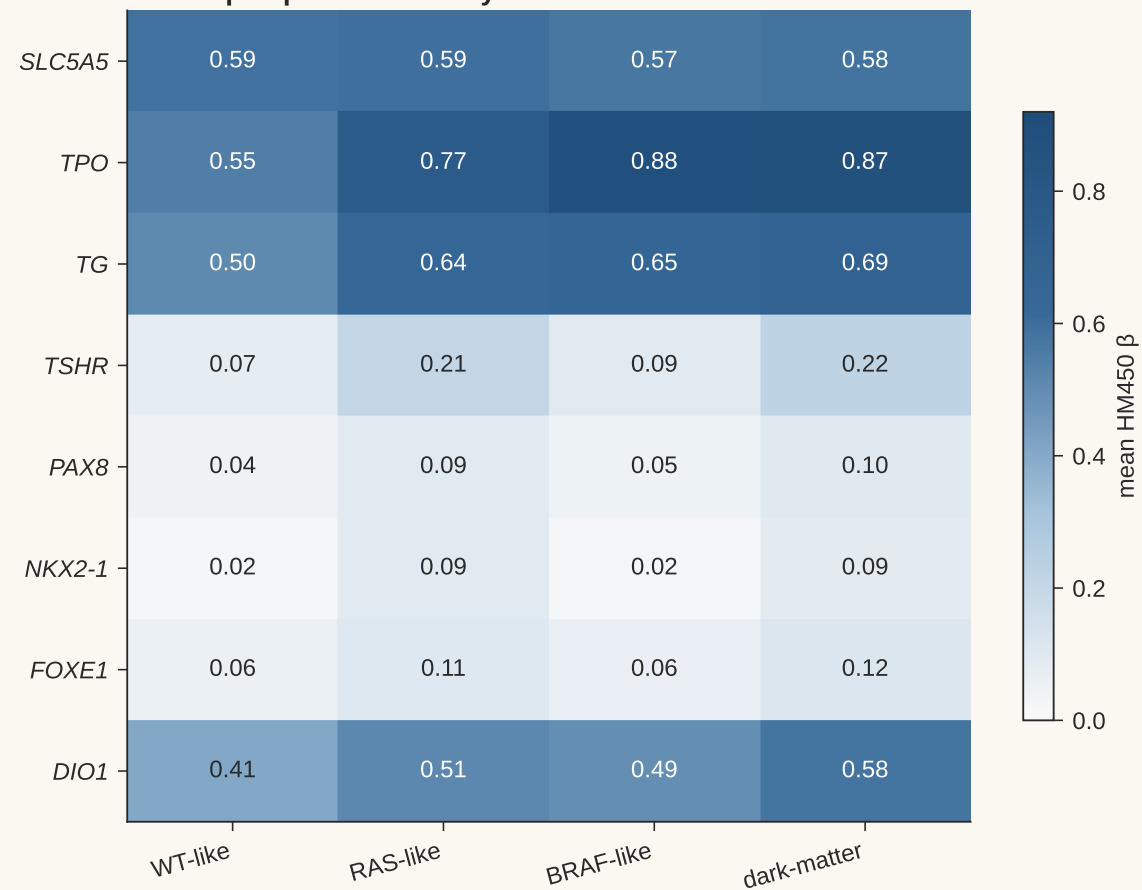


HM450 methylation β and RNA z per panel gene $\times$ R17 zone (TCGA-THCA, n = 518)

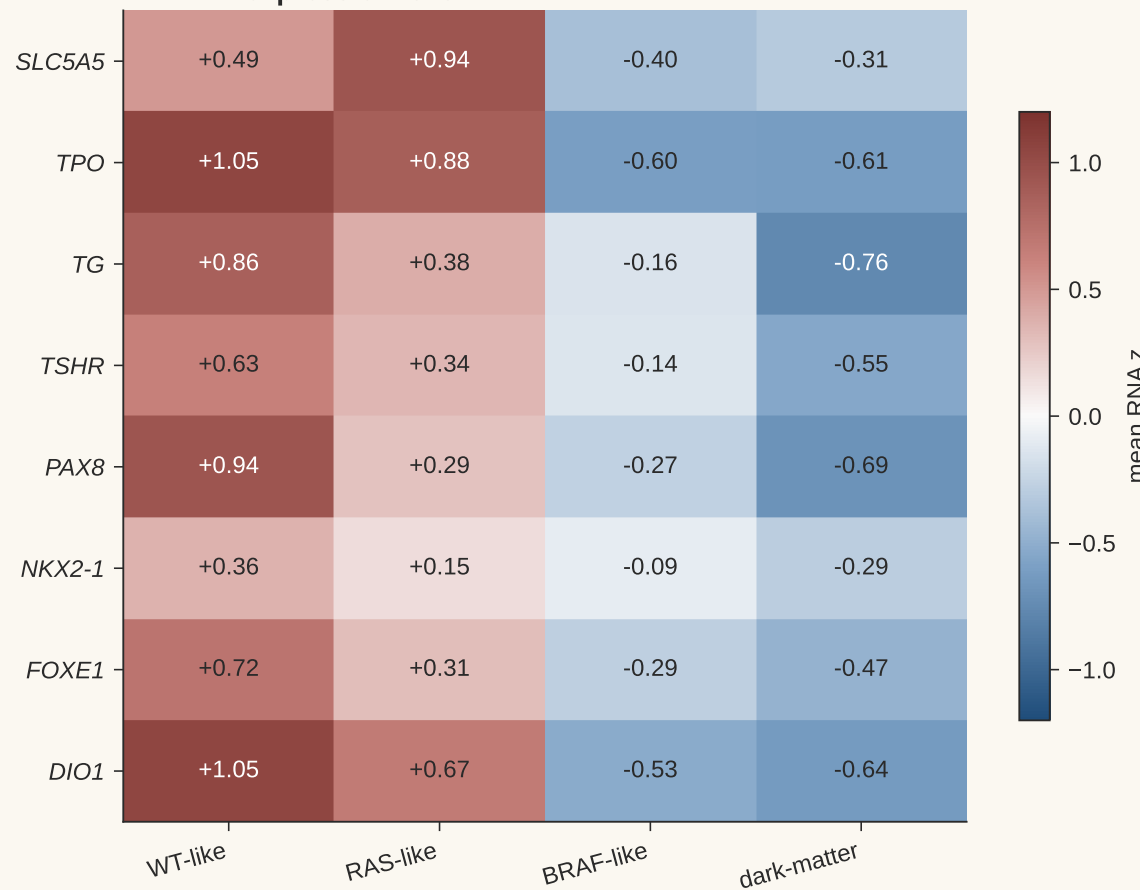
a

HM450 β · promoter methylation



b

RNA z · expression level



Dark-matter zone shows the highest mean β and the lowest mean RNA z across all panel genes — promoter hypermethylation directly tracks the silencing seen at the transcript level.